

ASSIGNMENT – 2

2.) Exploring the Table:

```
10  ~/
11
12  -- 2.) Exploring the Table
13  USE SuperstoreDB;
14  GO
15  SELECT TOP 10 *
16  FROM SuperStore;
17
18  -- Checking total rows
19  SELECT COUNT(*) AS Total_Rows
20  FROM SuperStore;
21
```

150 % No issues found Ln: 113, Ch: 1 MIXED CRLF UTF-8

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Some
2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Delux
3	CA-2016-136688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhes
4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CF
5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold
6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expr
7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322
8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 I
9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle
10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C2

Total_Rows
1 9994

3.) Apply WHERE Filters (Region, Category, Date, Sales):

Region:

```
20  -- 3.) Apply WHERE Filters
21  -- Region
22  SELECT *
23  FROM SuperStore
24  WHERE Region = 'West';
25
```

133 % No issues found Ln: 25, Ch: 23 TABS CRLF UTF-8

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name
3	CA-2016-136688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters t
6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Acc
7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322
8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone
9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by
10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge
11	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables
12	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal blac
14	CA-2016-161389	2016-12-05	2016-12-10	Standard Class	BA-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machinr
18	CA-2014-167164	2014-05-13	2014-05-15	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10000107	Office Supplies	Storage	Fellowes Super StorDrawer
19	CA-2014-143336	2014-08-27	2014-09-01	Second Class	ZD-21925	Zuschuss Dona...	Consumer	United States	San Franci...	California	94109	West	OFF-AR-10003056	Office Supplies	Art	Newell 341
20	CA-2014-143336	2014-08-27	2014-09-01	Second Class	ZD-21925	Zuschuss Dona...	Consumer	United States	San Franci...	California	94109	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone
21	CA-2014-143336	2014-08-27	2014-09-01	Second Class	ZD-21925	Zuschuss Dona...	Consumer	United States	San Franci...	California	94109	West	OFF-BI-10002215	Office Supplies	Binders	Wilson Jones Hanging View Binder, White, 1"
25	CA-2015-106320	2015-09-25	2015-09-30	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Tab
26	CA-2016-121755	2016-01-16	2016-01-20	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001634	Office Supplies	Binders	Wilson Jones Active Use Binders
27	CA-2016-121755	2016-01-16	2016-01-20	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories	Imation 8GB Mini TravelDrive USB 2.0 Flash D
43	CA-2016-101343	2016-07-17	2016-07-22	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10003479	Office Supplies	Storage	Eldon Base for stackable storage shelf, platinu

Query executed successfully. LAKSHAYSHARMAPC\SQLEXPRESSO... LAKSHAYSHARMAPC\laksh... SuperstoreDB 00:00:00 Row: 1, Col: 1 3,204 rows

Category:

26		
27	-- Category	
28		
29	SELECT *	
30	FROM SuperStore	
31	WHERE Category = 'Technology';	
32		
133 % No issues found Ln: 27, Ch: 12 TABS CRLF UTF-8		
Results Messages		
1	Row_ID	Order_ID
2	8	CA-2014-115812
3	12	CA-2014-115812
4	20	CA-2014-143336
5	27	CA-2016-121755
6	36	CA-2016-117590
7	41	CA-2015-117415
8	42	CA-2017-120999
9	45	CA-2016-118255
10	48	CA-2016-169194
11	55	CA-2016-105816
12	60	CA-2016-111682
13	63	CA-2015-135545
14	69	CA-2014-106376
15	87	CA-2017-155558
16	91	CA-2016-109906
17	101	CA-2016-155568
18	104	US-2015-156687
19	107	CA-2017-119004
20	108	CA-2017-119004
Ship_Date		
Ship_Mode		
Customer_ID		
Customer_Name		
Segment		
Country		
City		
State		
Postal_Code		
Region		
Product_ID		
Category		
Sub_Category		
Product_Name		
Mtel 5320 IP Phone VoIP phone		
Konfel 250 Conference phone - Charcoa		
Cisco SPA 501G IP Phone		
Imation 8GB Mini TravelDrive USB 2.0 Fi		
GE 30524EE4		
Panasonic Kx-TS550		
Verbatim 25 GB 6x Blu-ray Single Layer F		
Imation 8gb Micro Traveldrive Uab 2.0 Fi		
LF Elite 3D Dazzle Designer Hard Case C		
AT&T CL83451 4-Handset Telephone		
Imation 8gb Micro Traveldrive Uab 2.0 Fi		
Verbatim 25 GB 6x Blu-ray Single Layer F		
netTALK DUO VoIP Telephone Service		
Logitech LS21 Speaker System - PC Mult		
Panasonic Kx-TS550		
SanDisk Ultra 64 GB MicroSDHC Class 1		
Logitech K350 2.4Ghz Wireless Keyboard		
Memorex Mini Travel Drive 8 GB USB 2.0		
SanDisk Professional Endurance Plus Case		

Date:

32		
33	-- Date	
34		
35	SELECT *	
36	FROM SuperStore	
37	WHERE Order_Date BETWEEN '2016-01-01' AND '2016-11-7';	
38		
39		
40		
133 % No issues found Ln: 25, Ch: 26 TABS CRLF UTF-8		
Results Messages		
1	Row_ID	Order_ID
2	26	CA-2016-138888
3	27	CA-2015-121755
4	43	CA-2016-101343
5	45	CA-2016-118255
6	46	CA-2016-118255
7	48	CA-2016-169194
8	49	CA-2016-169194
9	56	CA-2016-111682
10	57	CA-2016-111682
11	58	CA-2016-111682
12	59	CA-2016-111682
13	60	CA-2016-111682
14	61	CA-2016-111682
15	62	CA-2016-111682
16	70	CA-2016-119823
17	71	CA-2016-106075
18	80	CA-2016-127208
19	81	CA-2016-127208
20	86	CA-2016-146896
Ship_Date		
Ship_Mode		
Customer_ID		
Customer_Name		
Segment		
Country		
City		
State		
Postal_Code		
Region		
Product_ID		
Category		
Sub_Category		
Product_Name		
Self-Adhesive Address Labels for Typewrit		
Wilson Jones Active Use Binders		
Imation 8GB Mini TravelDrive USB 2.0 Fi		
Eldon Base for stackable storage shelf, pl		
Verbatim 25 GB 6x Blu-ray Single Layer R		
Wilson Jones Leather-Like Binders with D		
Imation 8gb Micro Traveldrive Uab 2.0 Fi		
LF Elite 3D Dazzle Designer Hard Case C		
HomeOffice Personal File Carts		
Xerox 232		
Novimex Turbo Task Chair		
Array Parchment Paper, Assorted Colors		
Imation 8gb Micro Traveldrive Uab 2.0 Fi		
Plastic Binding Combs		
Prang Dustless Chalk Sticks		
Snap-A-Way Black Print Carbonless Rulec		
Avery Binding System Hidden Tab Execut		
1.7 Cubic Foot Compact "Cube" Office Ref		
Avery Heavy-Duty EZD Binder With Lockin		
Eldon Portable Mobile Mannequin		

Sales:

```
40  -- Sales
41  SELECT *
42  FROM SuperStore
43  WHERE Sales > 1000;
44
```

133 % ● No issues found Ln: 43, Ch: 19 TABS CRLF UTF-8

Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	Sales	Qua
2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.1840000000	9
2015-09-30	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables	Breford CR4500 Series Slim Rectangular Table	1044.6300000000	3
2015-09-21	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palais Royal Lawyers Bookcase, Royale Cherry	3083.4300000000	7
2016-12-10	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones	GE 30524EE4	1097.5440000000	7
2016-12-17	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City	New York	10024	East	TEC-PH-10002447	Technology	Phones	AT&T GL3451 4-Handled Telephone	1029.9500000000	5
2014-12-10	Standard Class	BS-11890	Brenden Sneed	Corporate	United States	Gilbert	Arizona	85234	West	OFF-AR-10002671	Office Supplies	Art	Hunt BOSTON Model 1606 High-Volume Electric Pencil S...	1113.3240000000	8
2016-12-09	Standard Class	JE-16165	Justin Ellison	Corporate	United States	Franklin	Wisconsin	53132	Central	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Round...	1991.9400000000	8
2014-09-12	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central	TEC-MA-10000822	Technology	Machines	Lexmark MX611dhe Monochrome Laser Printer	8159.9520000000	8
2014-09-12	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central	FUR-CH-10000827	Furniture	Chairs	SAFCO Arco Folding Chair	1740.0600000000	9
2015-01-09	Standard Class	VW-21775	Victoria Wilson	Corporate	United States	Medina	Ohio	44256	East	TEC-MA-10000864	Technology	Machines	Cisco 9971 IP Video Phone Charcoal	1188.0000000000	9
2014-06-06	Second Class	DW-13480	Dianna Wilson	Home Office	United States	Lakeville	Minnesota	55044	Central	FUR-CH-10004063	Furniture	Chairs	Global Deluxe High-Back Manager's Chair	2001.8600000000	7
2014-06-06	Second Class	DW-13480	Dianna Wilson	Home Office	United States	Lakeville	Minnesota	55044	Central	OFF-AP-10002945	Office Supplies	Appliances	Honeywell Enviracaire Portable HEPA Air Cleaner for 17' x...	1503.2500000000	5
2016-09-17	Standard Class	KC-16540	Kelly Collier	Consumer	United States	San Diego	California	92037	West	TEC-AC-10003832	Technology	Accessories	Logitech P710e Mobile Speakerphone	3347.3700000000	13
2014-09-21	Second Class	SB-20290	Sean Braxton	Corporate	United States	Houston	Texas	77036	Central	TEC-MA-10000622	Technology	Machines	Lexmark MX611dhe Monochrome Laser Printer	3059.9620000000	3
2014-09-21	Second Class	SB-20290	Sean Braxton	Corporate	United States	Houston	Texas	77036	Central	TEC-MA-10003353	Technology	Machines	Xerox WorkCentre 6505DN Laser Multifunction Printer	2519.9500000000	7
2015-09-28	Second Class	SC-20725	Steven Cartwright	Consumer	United States	Houston	Texas	77070	Central	TEC-PH-10001760	Technology	Phones	Bose SoundLink Bluetooth Speaker	1114.4000000000	7
2015-11-06	Standard Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	90004	West	FUR-TA-10002903	Furniture	Tables	Bevis Round Bullnose 29" High Table Top	1038.8400000000	5
2016-11-03	Standard Class	JD-15895	Jonathan Doherty	Corporate	United States	Belleville	New Jersey	7109	East	OFF-ST-10002756	Office Supplies	Storage	Tennisco Stur-D-Stor Boltless Shelving, 5 Shelves, 24" Dee...	1082.4800000000	8
2014-11-09	Standard Class	NM-18445	Nathan Matutz	Home Office	United States	New York City	New York	10024	East	TEC-MA-10002927	Technology	Machines	Canon imageCLASS MF7460 Monochrome Digital Laser ...	3991.9800000000	2
2016-10-01	First Class	AR-10060	Adam Rallapureddy	Home Office	United States	New York City	New York	10009	East	OFF-RL-10004905	Office Supplies	Binders	GRN PowerRound P400 Electric Roundbin System	4145.1680000000	4

4.) Aggregation Using GROUP BY (Sales, Quantity, Average):

```
SQLQuery1.s...laksh (55))*
46  -- 4.) Aggregation Using GROUP BY
47
48  -- Total Sales by Region
49  SELECT Region,
50  SUM(Sales) AS Total_Sales
51  FROM SuperStore
52  GROUP BY Region;
53
54
55  -- Total Quantity by Category
56  SELECT Category,
57  SUM(Quantity) AS Total_Quantity
58  FROM SuperStore
59  GROUP BY Category;
60
61  -- Average Sales by Category
62  SELECT Category,
63  AVG(Sales) AS Average_Sales
64  FROM SuperStore
65  GROUP BY Category;
66
```

83 % ● No issues found

Results	Messages
2	South 391721.9050000000
3	West 725457.8245000000
4	Central 501239.8908000000

Category	Total_Quantity
1	Office Supplies 22906
2	Furniture 8028
3	Technology 6939

Category	Average_Sales
1	Office Supplies 119.3241008961
2	Furniture 349.8348869872
3	Technology 452.7092761234

5.) Sort and Limit Results

```
68
69  -- 5.) Sort and Limit Results
70
71  -- Top 10 Products
72  SELECT TOP 10 Product_Name,
73             SUM(Sales) AS Total_Sales
74  FROM SuperStore
75  GROUP BY Product_Name
76  ORDER BY Total_Sales DESC;
77
78  -- Top Categories
79  SELECT Category,
80             SUM(Sales) AS Total_Sales
81  FROM SuperStore
82  GROUP BY Category
83  ORDER BY Total_Sales DESC;
```

100 % No issues found

Results Messages

	Product_Name	Total_Sales
1	Canon imageCLASS 2200 Advanced Copier	61599.82400000...
2	Fellowes PB500 Electric Punch Plastic Comb Binding ...	27453.38400000...
3	Cisco TelePresence System EX90 Videoconferencing ...	22638.48000000...
4	HON 5400 Series Task Chairs for Big and Tall	21870.57600000...
5	GBC DocuBind TL300 Electric Binding System	19823.47900000...
6	GBC Ibimaster 500 Manual ProClick Binding System	19024.50000000...
7	Hewlett Packard LaserJet 3310 Copier	18839.68600000...
8	HP Designjet T520 Inkjet Large Format Printer - 24" C...	18374.89500000...
9	GBC DocuBind P400 Electric Binding System	17965.06800000...
10	High Speed Automatic Electric Letter Opener	17030.31200000...

	Category	Total_Sales
1	Technology	836154.0330000...
2	Furniture	741999.7953000...
3	Office Supplies	719047.0320000...

6.) Use Cases:

SQLQuery1.s...laksh (55))* ✕

```
--  
86  -- 6.) Use Cases  
87  
88  -- Monthly Sales Trend  
89  SELECT YEAR(Order_Date) AS Year,  
90         MONTH(Order_Date) AS Month,  
91         SUM(Sales) AS Total_Sales  
92  FROM SuperStore  
93  GROUP BY YEAR(Order_Date), MONTH(Order_Date)  
94  ORDER BY Year, Month;  
95
```

100 % ✓ No issues found

Results Messages

	Year	Month	Total_Sales
1	2014	1	14236.8950000000
2	2014	2	4519.8920000000
3	2014	3	55691.0090000000
4	2014	4	28295.3450000000
5	2014	5	23648.2870000000
6	2014	6	34595.1276000000
7	2014	7	33946.3930000000
8	2014	8	27909.4685000000
9	2014	9	81777.3508000000
10	2014	10	31453.3930000000
11	2014	11	78628.7167000000
12	2014	12	69545.6205000000
13	2015	1	18174.0756000000
14	2015	2	11951.4110000000
15	2015	3	38726.2520000000

```

96  -- Top 10 Customers
97  SELECT TOP 10 Customer_Name,
98         SUM(Sales) AS Total_Sales
99  FROM SuperStore
100 GROUP BY Customer_Name
101 ORDER BY Total_Sales DESC;
102
103  -- Duplicate Values
104  SELECT Order_ID,
105         COUNT(*) AS Duplicate_Count
106  FROM SuperStore
107  GROUP BY Order_ID
108  HAVING COUNT(*) > 1;

```

83 %   No issues found

 Results  Messages

	Customer_Name	Total_Sales
1	Sean Miller	25043.050000...
2	Tamara Chand	19052.218000...
3	Raymond Buch	15117.339000...
4	Tom Ashbrook	14595.620000...
5	Adrian Barton	14473.571000...
6	Ken Lonsdale	14175.229000...
7	Sanjit Chand	14142.334000...
8	Hunter Lopez	12873.298000...
9	Sanjit Engle	12209.438000...
10	Christopher Con...	12129.072000...

	Order_ID	Duplicate_Count
1	CA-2017-143...	3
2	CA-2017-125...	2
3	CA-2014-151...	3
4	CA-2015-130...	3
5	CA-2017-168...	4
6	US-2014-112...	3
7	CA-2017-165...	3
8	CA-2014-116...	2

7.) Validate Results:

```
113 -- 7.) Validate Results
114 SELECT COUNT(*) AS Total_Rows
115 FROM SuperStore;
116
117 -- Check missing values
118 SELECT *
119 FROM SuperStore
120 WHERE Sales IS NULL
121        OR Quantity IS NULL
122        OR Customer_Name IS NULL
123        OR Region IS NULL;
124
125 -- Count Missing Values
126 SELECT
127 SUM(CASE WHEN Sales IS NULL THEN 1 ELSE 0 END) AS Missing_Sales,
128 SUM(CASE WHEN Quantity IS NULL THEN 1 ELSE 0 END) AS Missing_Quantity,
129 SUM(CASE WHEN Customer_Name IS NULL THEN 1 ELSE 0 END) AS Missing_Customers
130 FROM SuperStore;
131
```

93 % ✓ No issues found

Ln: 131, Ch: 17 MIXED CRF UTF-8

Results Messages

Total_Rows
1 9994

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	Sales	Quantity	Discount	Profit
--------	----------	------------	-----------	-----------	-------------	---------------	---------	---------	------	-------	-------------	--------	------------	----------	--------------	--------------	-------	----------	----------	--------

Missing_Sales	Missing_Quantity	Missing_Customers
1 0	0	0